

**Peeping beyond the pond: transferability of a thermal threshold for spring peeper calling
onset, and the limits of large-scale call surveys**

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Introduction

As climate change progresses, it continues to pose a critical threat to biodiversity. In cases where organism behaviours are tied to climatic conditions, changes to cues risk phenological mismatches (Visser et al., 2012). Because ectotherms depend on external heat to reach the physiological thresholds that govern development and activity, their life-history timing is tightly coupled to environmental conditions (Jarošík et al., 2004). In particular, amphibian breeding behaviours have significantly advanced in response to increased warming (Parmesan, 2007). Anuran calling has shifted earlier in response to both decadal climate warming (Gibbs & Breisch, 2001), and within-season fluctuations such as false springs (Ethier et al., 2026). Because anuran calls are distinct and are readily observable, their calling events serve as opportunities for tracking climate change, modelling current and historical breeding events, and predicting future behaviours. However, to model calling behaviour and its relationship to temperature shifts, we first need to identify which specific temperature cues an organism is responsive to.

Temperature can be measured by various metrics spanning different time periods, and species or populations may respond differently to each one, depending on the specific behaviour being cued (Klaus & Loughheed, 2013; Semlitsch, 1985). While measures of temperature are correlated, identifying the specific cues that an organism tracks to trigger a change in phenotype gives us the ability to predict when onset will occur more precisely. For instance, if two candidate cues predict the same calling onset date under historical conditions, observation alone cannot distinguish between them; but when they diverge under new conditions, the precision of the prediction matters. Distinguishing among candidate cues requires observations across sites that differ thermally, since cues that coincide at one location may diverge across a species' range.

Spring peepers are a useful study system because their range spans from south- to northeastern North America across approximately 26° latitude (IUCN SSC Amphibian Specialist Group, 2022), providing distinct regions to differentiate sensitivity to candidate thermal cues. Additionally, their distinctive calls make them easily identifiable by non-specialists. Pinpointing the date of first call (DFC), however, requires frequent observation. The North American Amphibian Monitoring Program (NAAMP), a United States Geological Survey effort, produced frog calling observations across the eastern and central United States (Foreman et al., 2017), providing consistent data that can be combined with climate records to model onset beyond a single site.

In a test of several climate indices, Lovett identified a thermal accumulation index as the best predictor of spring peeper calling onset at a single pond in New York (2013). The index began accumulating on February 1st, used a base temperature of 3°C, and reached a threshold of 44.3 degree days (TS3). When applied to historical climate records, his model indicated that spring peeper calling onset had advanced approximately 11 days since 1949. His work identifies TS3 as a potential cue that the species responds to; however, it is unclear whether this thermal index transfers beyond the single site that Lovett tests and potentially acts as a physiological constant, or if it is a feature of the specific population at Lovett's pond.

Here, I (1) investigate whether the TS3 cue predicts observed spring peeper DFC at multiple sites across the species range (Figure 1); (2) determine whether TS3 retains explanatory power beyond a site's geographic position in a multiple regression framework; and (3) quantify the resolution with which existing monitoring designs capture calling onset in an early-breeding anuran, and identify designs that would improve it. If the TS3 threshold represents a species-level physiological requirement rather than a site-specific fit, predicted DFC should track

observed onset with a slope near 1 and an intercept near 0, and should remain a significant predictor after geographic position is accounted for.

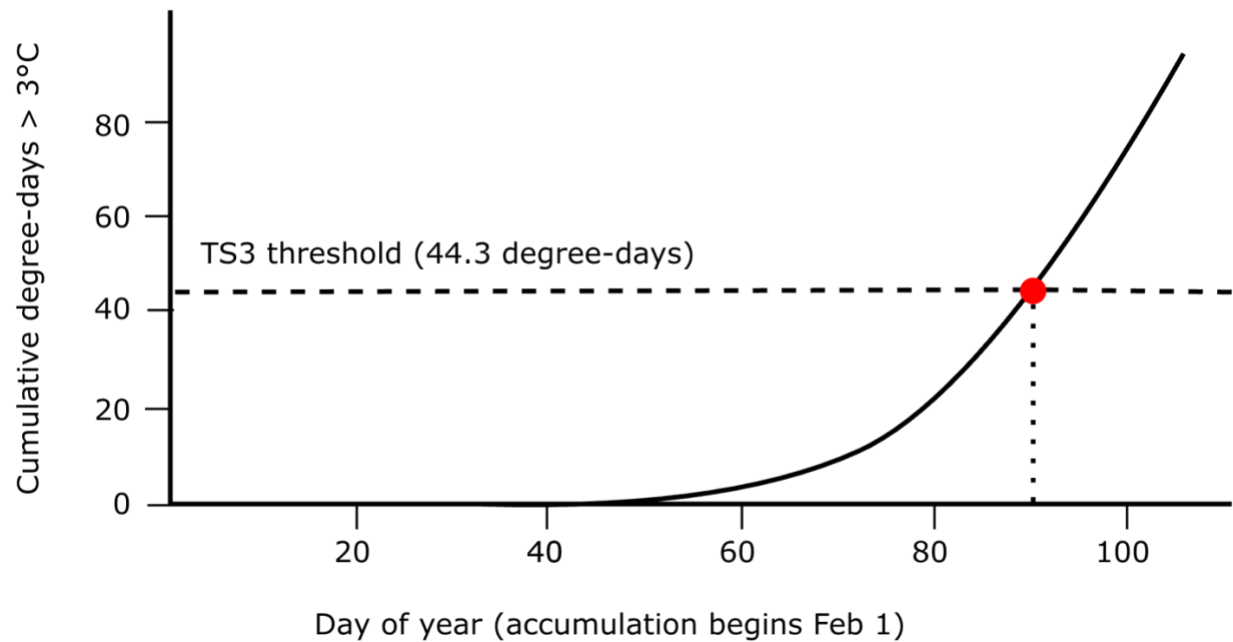


Figure 1. Graphical representation of Lovett's TS3 model. Dashed line represents the TS3 threshold of 44.3 growing degree days at a base of 3°C, and dotted line represents the day of year which the threshold was crossed. The window for this model starts at February 1st, or Julian day 32. Here, predicted DFC is 90.

Methods

Workflow

This study was conducted in three parts (Figure 2). The first phase involved cleaning and censoring the North American Amphibian Monitoring Program (NAAMP) observational data to prepare for spring peeper calling onset modelling. The second phase was the implementation of Lovett's (2013) TS3 model through raster computation of climate data from GEE. The final phase combined the computed predicted onset dates based on TS3 for each route represented in the NAAMP observations for modelling observed calling onset.

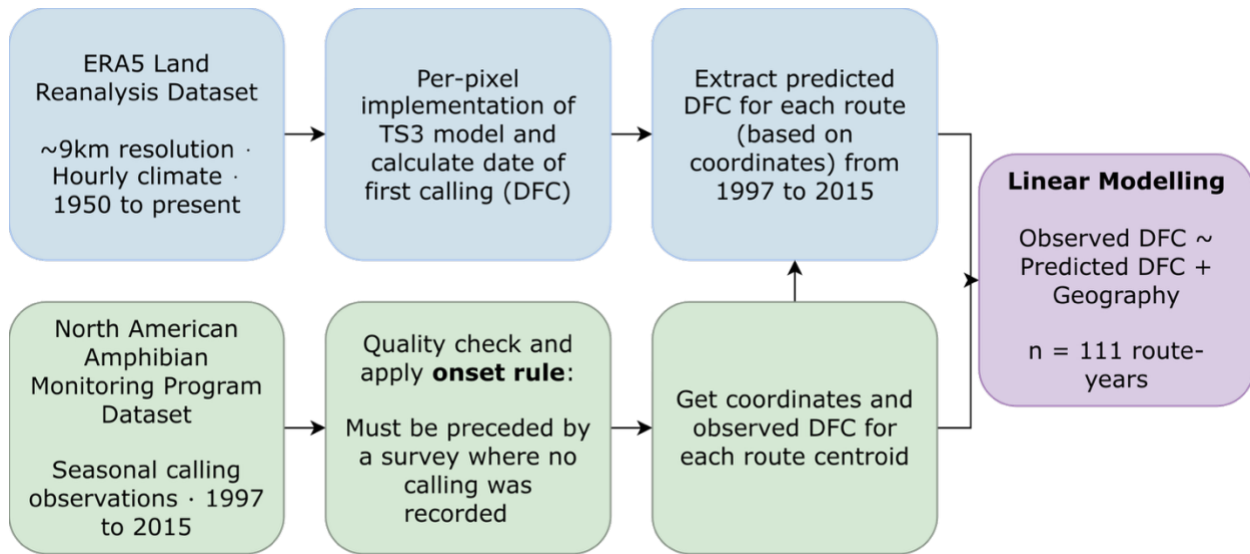


Figure 2. Modelling workflow. Climate data processed in Google Earth Engine (top, blue) and NAAMP survey data quality-controlled processed in R (bottom, green) were combined to model observed DFC as a function of TS3 and geography (middle, purple). Note that Daymet 1-km resolution validation was conducted using the same process as the ERA5 Land Reanalysis dataset.

Computational Resources

All analyses were conducted on an Apple MacBook Pro (M4 architecture). The Google Earth Engine (GEE) IDE was used for climate data calculations with an account registered for non-commercial use. Git was used for version control.

Data Sources

The ERA5 Land Reanalysis dataset (Muñoz Sabater, 2019) was accessed through GEE. It contains hourly climate data from 1950 to 5-days behind real-time with a resolution of 9 km per pixel across the globe. Each pixel is approximately 81 km². The `temperature\_2m` band was the selected band for primary temperature calculations.

The Daymet dataset (version 4; Thornton et al., 2022) was also accessed through GEE. At the time of the analysis, the dataset contained daily climate summaries from 1980 to 2025 with a 1 km per pixel resolution across continental North America, Hawaii, and Puerto Rico. Each pixel

is approximately 1 km². The `tmax` and `tmin` bands were selected for validation of the ERA5-derived temperature calculations.

The NAAMP data (Foreman, Grant and Weir, 2017) was accessed from the United States Geological Survey (USGS) and includes observational data generated by citizen science efforts across 26 states. The dataset tracks amphibian calling trends collected across a network of random and non-random roadside routes with listening stops located near wetlands, following a standardized protocol. Data collection at each stop included frog and toad occupancy, calling intensity, and environmental data. This analysis uses the raw, verified NAAMP intensity records spanning 1994 to 2015.

Cleaning NAAMP Data

The NAAMP protocol had surveyors record positive detections only, so non-detections were inferred by constructing the full survey grid of `Route` (site) by `Year` by `Run` (survey) by `Species` and assigning a `Calling Index` (intensity of calling) of zero where a species was not recorded. This recovered non-detections only for runs on which at least one species was recorded; surveys where no calling of any species was detected generated no record and could not be recovered. Onset was defined as the first survey in a season at which the calling index reached a score of 1 or greater, provided the preceding survey in that season recorded no calling. This requirement ensured the transition from silence to calling was directly observed rather than inferred. Route-years in which the interval between the bracketing surveys exceeded 30 days were excluded. Following data cleaning and quality control, 90 routes over 19 years (1997 to 2015) remained, for a total of 111 route-years. Coordinates for each route were taken as the mean of the `Stop` (within-route replication) coordinates within each route and fed directly into GEE to extract temperature data.

Implementing Lovett Model in GEE

Lovett's (2013) TS3 index was implemented as a per-pixel raster computation in GEE. For each year from 1997 to 2015, hourly `temperature\_2m` values were averaged to daily means and converted from °K to °C. Daily degree-days were calculated as the difference between daily mean temperature and a base of 3°C, with negative values set to zero. Degree-days were accumulated from February 1st (Julian day 32) across a 150-day search window, and the predicted date of first calling (DFC) was defined as the first day on which the cumulative sum reached 44.3 degree-days. Pixels not reaching the threshold within the window were masked. There were no predicted DFC dates that were exactly DOY 32 or 150, indicating the search window was wide enough, and did not censor any predictions.

Predicted DFC was extracted at NAAMP route coordinates using `reduceRegions`, yielding one predicted value per route per year. Extractions were run in multi-year chunks to remain within GEE memory limits.

To assess whether the coarse resolution of ERA5-Land influenced results, the identical procedure was repeated using Daymet. As Daymet provides daily `tmax` and `tmin` rather than hourly values, daily mean temperature was calculated as the average of the two.

Modelling DFC

Observed onset was modelled as a function of predicted DFC using linear models in R (version 4.5.1; R Core Team, 2025). The tidyverse (version 2.0.0; Wickham et al., 2019), lubridate (version 1.9.5; Grolemund & Wickham, 2011), and lme4 (version 2.0-1; Bates et al., 2015) packages were used for data cleaning and manipulation, date handling, and modelling, respectively. Latitude and longitude were centred on their means, and survey year was evaluated

as both a fixed effect and a random intercept; a route-level random intercept was not identifiable, as 74 of 90 routes (82%) contributed a single observation. Candidate models were compared using AIC and nested F-tests, with variance inflation factors assessed for collinearity and Cook's distance for influential observations.

Code Availability

The computational workflows and analysis scripts used to generate the results in this study are publicly available on GitHub at <https://github.com/farah-y-sadoon/predicted-peeper-calling-onset.git>.

Results

Applying the onset criteria to NAAMP records for spring peepers yielded 111 route-years across 90 routes and 19 survey years (1997 to 2015), spanning 34.5 to 45.1°N. Most routes contributed a single observation (74 of 90, 82%), with the remainder contributing two or three. Observed onset ranged from day 27 to day 190, with predicted DFC ranging from day 35 to day 126 (Figure 3). Only 316 of 7,678 peeper route-years (4%) included a survey recording silence before the DFC, and of these, 53% recorded full chorus at that first detection. Because chorus intensity builds over days, calling had typically begun before the recording survey. After excluding route-years where the survey intervals exceeded 30 days, 113 remained. Two route-years did not have documented coordinates, which excluded them from the analysis.

The initial model with TS3 as a sole predictor saw approximately 17% of the variance in observed calling onset explained ($R^2 = 0.177$, $p < 0.001$). While the fitted slope did not differ significantly from the 1:1 expectation ($\beta - 1 = -0.29$, $SE = 0.15$, $p = 0.055$), the intercept was

significantly positive (72.6 days, $p < 0.001$), indicating that observed onset was offset later than predicted by a roughly constant amount rather than scaled differently (Figure 3, Table 1).

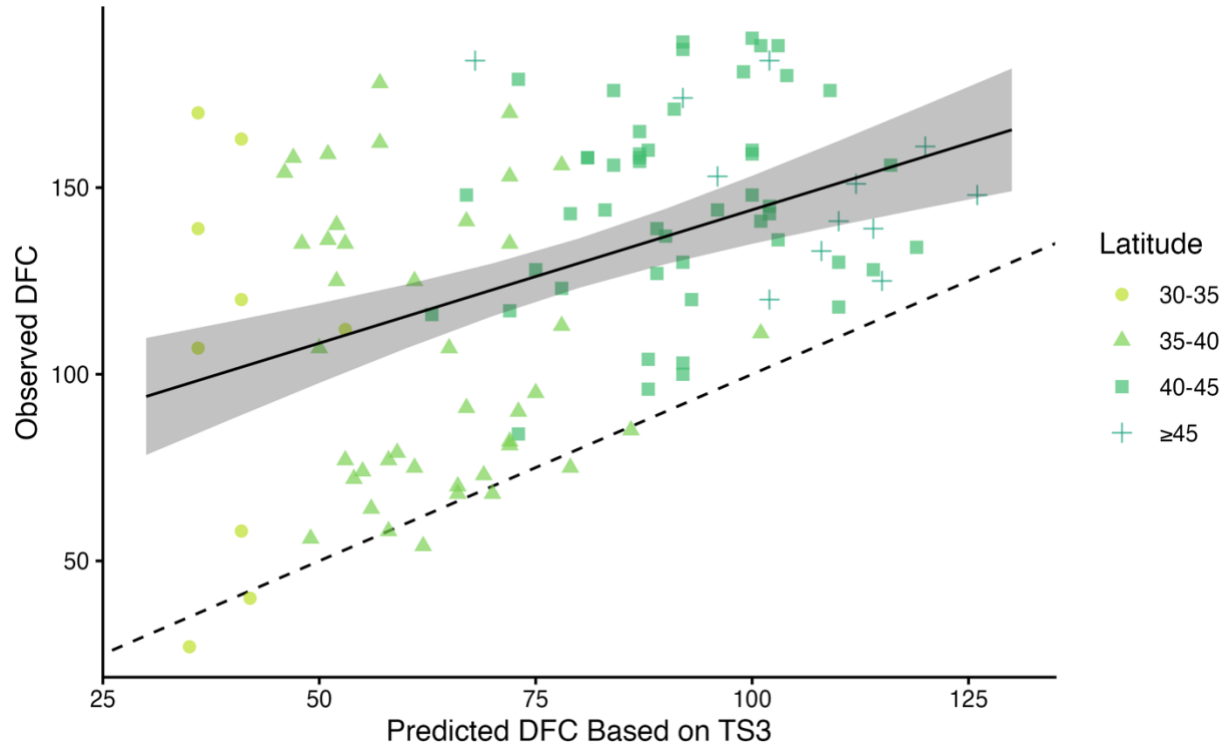


Figure 3. Observed calling onset against TS3-predicted onset at 90 NAAMP routes (111 route-years) by latitude band. TS3 significantly predicted observed onset (slope = 0.71, $p < 0.001$), but observations occurred systematically later than predicted (intercept = 72.6 days, $p < 0.001$). The dashed line indicates 1:1 agreement. Fitted slope was not significantly different from the 1:1 line ($\beta - 1 = -0.29$, SE = 0.15, $p = 0.055$), but slight convergence in 1:1 line and slope is present.

Survey year appeared to predict onset when fitted alongside TS3 (-1.88 days per year, $p = 0.005$) but was non-significant once latitude and longitude were included ($p = 0.93$). Fitted as a random intercept, year accounted for 13% of total variance, but this variance component had a zero estimate when geographic predictors were included. Both results indicate that the temporal trend likely reflects variation in which routes were surveyed in which years rather than a shift in calling phenology.

Table 1. Comparison of candidate models predicting observed calling onset. Geography alone outperformed TS3 alone ($\Delta AIC = 27.2$) and adding TS3 to geography produced a small improvement ($F_{1, 107} = 5.85$, $p = 0.017$). ΔAIC is relative to the best-supported model (*).

<i>Model</i>	<i>k</i>	<i>R</i> ²	<i>RSE</i>	<i>ΔAIC</i>
TS3	2	0.177	35.1	31.1
TS3 + year	3	0.234	34.0	25.2
Latitude + Longitude	3	0.368	30.9	3.91
TS3 + Latitude + Longitude*	4	0.400	30.2	0
TS3 x Latitude + Longitude	5	0.400	30.4	1.97

Geographic position alone explained more variance in observed calling onset than TS3 ($R^2 = 0.368$ vs $R^2 = 0.177$) and was better supported by AIC ($\Delta AIC = 27.2$). The best supported model included TS3, latitude and longitude ($R^2 = 0.400$, Table 1). Adding TS3 to the geographic predictors improved fit ($F_{1, 107} = 5.85$, $p = 0.017$), but its coefficient was negative (-0.603 , Table 2, Figure 4), which is unexpected given the mechanism of the thermal sum model. Holding TS3 and longitude constant, onset occurred approximately 9.6 days later per degree of latitude north ($p < 0.001$; Table 2).

Table 2. Coefficients from the best-supported model (observed DFC ~ TS3 + latitude + longitude; $R^2 = 0.400$, $n = 111$ route-years). Latitude and longitude are centred on their means. Holding TS3 constant, onset occurred approximately 9.6 days later per degree of latitude north. Holding location constant, each additional day of delay in reaching the TS3 threshold was associated with onset approximately 0.6 days earlier.

<i>Term</i>	<i>Estimate</i>	<i>SE</i>	<i>p</i>
Intercept	176	19.8	< 0.001
Predicted DFC based on TS3 (days per day)	-0.603	0.249	0.017
Latitude (days per °N)	9.64	1.56	< 0.001
Longitude (days per °E)	1.26	0.374	0.001

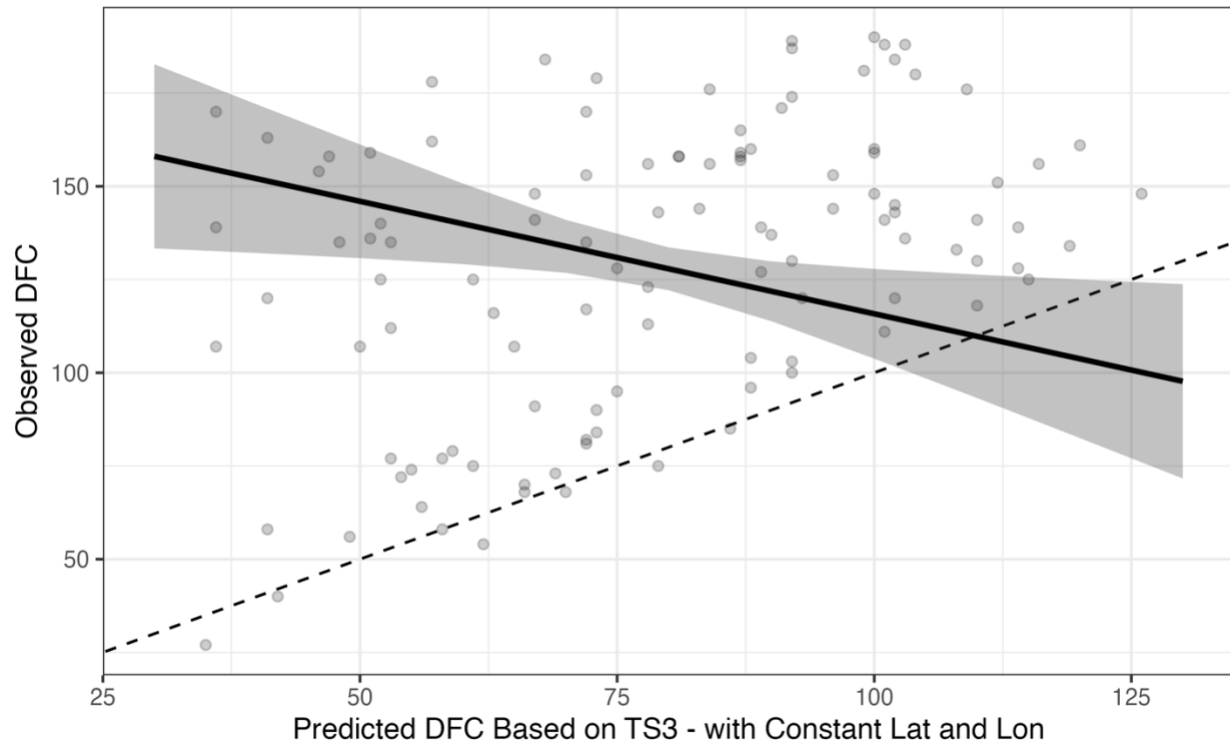


Figure 4. Observed calling onset against TS3-predicted onset, with latitude and longitude held at their mean values ($n = 111$ route-years). After accounting for geographic position, the fitted relationship is negative (-0.603 , $SE = 0.25$, $p = 0.017$), crossing the 1:1 line rather than converging toward it. Dashed line indicates 1:1 agreement; shaded band is the 95% CI.

The negative coefficient of TS3 was investigated further by looking at latitude observation bias, influential observations, collinearity, and climate data resolution. A sensitivity analysis was performed, removing all points from the lowest latitude bin (30° to 35°). Since lower latitudes are likely to reach the degree day accumulation faster than higher ones, a floor-effect was considered, where the observations from lower latitudes were creating bias in the model because they were calling before the prediction period started (DOY 32). The sensitivity analysis yielded a slight improvement in the model performance overall ($R^2 = 0.437$), but the TS3 estimate remained negative (-0.537). Influential observations were explored with Cook's distance, which found no points over the 1.0 cutoff ($\max = 0.085$), but found 5 influential points above the $4/n$ threshold. Removing the influential observations slightly improved the R^2 (0.511),

but the TS3 coefficient was more negative (-0.784), meaning that these observations were not the cause of the negative relationship between observed onset date and TS3. Next, collinearity assessed with variance inflation factors highlighted moderate correlations (TS3 = 3.852, centered latitude = 4.023, centered longitude = 1.131). Finally, the model was reconstructed using Daymet (Thornton et al., 2022) revealing a similar result to the TS3 + geography model ($R^2 = 0.393$, $\beta_{TS3} = -0.528$) using ERA5 Land (Muñoz Sabater, 2019).

Discussion

In this study, I investigated the TS3 cue identified by Lovett (2013) for spring peeper calling onset and its predictive power across 90 routes, and 111 route-years within the species range. I used GEE to spatially implement the TS3 model and extract the predicted DFC dates from both the ERA5 Land Reanalysis and Daymet datasets for consistent temperature measurements. Additionally, I explored whether TS3 has additional predictive power for the observed DFC at the represented route-years after accounting for geographic position. Finally, this analysis uses a USGS coordinated citizen science survey of spring peeper calling onset across 26 states. Although their survey was not designed specifically for detecting precise onset (Foreman et al., 2017), this analysis characterizes gaps in the survey protocol for modelling onset and what improvements can be made for future versions so that predictive modelling is more accurate.

When observed DFC was modelled as a function of the TS3 cue threshold, the slope did not deviate from the 1:1 expectation, but the intercept diverged substantially from 0 (Figure 3). The intercept was significantly positive (72.6 days, $p < 0.001$), corresponding to observed onset occurring roughly 45-50 days later than predicted at typical values of predicted DFC. This difference in onset may reflect survey timing, as the NAAMP protocol prioritized detection over capturing DFC (Foreman et al., 2017). The accumulation window for Lovett's model begins on

February 1st, which may have introduced some bias for the Northern routes, where the 3°C degree day accumulation would have accrued slower. Moreover, these results may indicate that the TS3 threshold does not transfer well beyond Lovett's pond, or that other environmental cues in combination with TS3 are required for accurate prediction of calling onset. In fact, Lovett found that a single year where his pond was dry notably shifted his results, where excluding the anomalous year resulted in 93% of variance in DFC explained by TS3 alone as opposed to 51% overall when all years were included (2013). This phenomenon has also been observed in other ephemeral pond breeders such as the common spadefoot toad (*Pelobates fuscus*), where dry conditions were associated with a delay in migration to breeding sites (Dalpasso et al., 2023) despite rising temperatures. As well, spring peeper emergence is highly associated with precipitation (Baker et al., 2024), which further supports that breeding behaviours like calling are contingent on precipitation patterns.

Geography alone explained more variance in calling onset of spring peepers than TS3 alone, though the best-supported model combined both (Table 1). Since most routes were represented by a single observation, variation in predicted DFC across the dataset was predominantly spatial rather than temporal. Lovett developed the TS3 threshold at a single location over 15 years (2013) where the geography was held constant and year-to-year thermal variation remained. In the spatial dataset analysed here, coordinates measure geographic position directly and TS3 does so indirectly, which is likely why geography is the predominant predictor here.

When TS3 and longitude were held constant, a one-degree increase in latitude was associated with an approximately 9-day delay in calling (Table 2). Because thermal accumulation is accounted for, this suggests that factors covarying with latitude, but not with accumulated

warmth, contribute to onset timing. A potential mechanism could be photoperiod, as day length varies with latitude but is unaffected by warming. In fact, Thompson and colleagues found that day of year predicts calling independently of temperature and rainfall in Australian frogs examined at continental scale (Thompson et al., 2022). Additionally, since winter conditions impact organisms through mechanisms different from ones operating in spring (Williams et al., 2015), temperature-dependent processes occurring before the accumulation window may shape onset in ways that TS3 would not capture. For instance, snowmelt timing strongly predicts spawning date in the common frog, with a 10-day delay in snowmelt corresponding to a 5.8-day delay in spawning; because snowmelt trends differed between elevations, breeding advanced at low-elevation sites but was delayed at high-elevation sites over the same period (Bison et al., 2021).

TS3 retained a negative slope when geographic predictors were held constant, indicating an inverse relationship between observed and predicted onset (Table 2, Figure 4). This is counterintuitive given that spring warming initiates breeding activity in early-breeding amphibians, and that thermal accumulation above 3°C predicted calling onset at Lovett's site but may be explained by other environmental factors such as false spring risk. Earlier environmental cues for spring peeper emergence are more frequently followed by subfreezing temperatures compared to historical conditions, linking earlier emergence to an increased risk in freezing (Baker et al., 2024). There was no significant interaction between latitude and TS3 in this study, so the relationship did not vary noticeably across the sampled routes. Whether false-spring dynamics account for the negative association cannot be determined here, as my analysis does not distinguish sites where early thermal accumulation was followed by freezing conditions from those where it was not.

Here, I investigated the effectiveness of the TS3 cue for predicting calling onset using data generated by NAAMP. Since the protocol prioritized capturing species presence during peak calling periods, the observations were not well suited for resolving DFC precisely. Obtaining reliable calling onset dates therefore excluded most of the available data: 4% of the spring peeper route-years included a prior survey recording no calling at the same route. These results point toward design features for a monitoring program aimed specifically at detecting calling onset. Visit frequency and the absence of recorded non-detections were the main reasons for data loss here, and both are addressable. Recording zeroes directly, rather than inferring them, removes the need to reconstruct survey grids and allows onset to be bounded between a documented silent visit and first calling visit. Beginning surveys before the earliest expected calling date would capture onset in early-breeding species that call before monitoring begins – here, the difference between observed and predicted onset dates was approximately 45-50 days, which may suggest an appropriate shift in timing for spring peepers. Repeated visits to fixed sites, rather than routes surveyed a few times per season, would build within-site replication for separating year-to-year thermal variation from site-to-site differences. Additionally, autonomous recording units offer continuous coverage without observer-timing bias and can be scored on the same calling index used by NAAMP (Ethier et al., 2026).

Overall, this study provides a framework for modelling anuran breeding phenology using climate data. Additionally, the results can be used to inform future monitoring programs and improve the accuracy and applicability of calling observations while addressing gaps in existing monitoring protocols. Modelling the breeding phenology of amphibians, such as the spring peeper, can support the development of more effective monitoring protocols and contribute to broader conservation efforts aimed at protecting and preserving biodiversity.

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